

Hanpu Liu

PH.D. CANDIDATE · PRINCETON UNIVERSITY

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Education

Princeton University

DOCTOR OF PHILOSOPHY, ASTROPHYSICS

New Jersey, USA

Aug 2024–present

Peking University (Yuanpei College)

BACHELOR OF SCIENCE, PHYSICS

China

Sep 2020–Jul 2024

University of California, Berkeley

EDUCATION ABROAD PROGRAM

California, USA

Aug 2022–Dec 2022

Research Experience

Spectral modeling of Little Red Dots (LRDs) (*HL+25,26*)

ADVISORS: DR. YAN-FEI JIANG, PROF. ELIOT QUATAERT, PROF. JENNY E. GREENE

- Proposed LRDs as super-Eddington accretion flows onto supermassive black holes
- Resolved UV-optical spectral puzzles of LRDs with radiative transfer physics
- Calculated model spectral library for LRDs
- Initiated a pathway to inferring the black hole mass based on spectral absorption lines

Princeton Univ., NJ

2025–Present

Formation of magnetic field structures in collisionless shocks (*HL+ in prep*)

ADVISOR: PROF. ANATOLY SPITKOVSKY

- Revealed the mechanism of large-scale magnetic field growth in shock simulations
- Recovered phenomena observed in shock simulations using idealized numerical setups
- Predicted long-term magnetic field evolution in γ -ray bursts

Princeton Univ., NJ

2024–Present

Multi-messenger Clues of Early Massive Black Hole (MBH) Evolution (*HL & Inayoshi 24*)

ADVISOR: PROF. KOHEI INAYOSHI

- Constrained MBH major merger rates from quasar statistics and nanohertz gravitational wave (GW) background signals
- Predicted space-based GW detection of individual merger events and the stochastic background
- Pinpointed millihertz GW events feasible for follow-up electromagnetic observation

Peking Univ., Beijing, China

2023–2024

Dust–Gas Instabilities in Protoplanetary Disks (*HL & Bai 23*)

ADVISOR: PROF. XUENING BAI

- Calculated unstable linear wave modes in turbulent dusty disk rings
- Discovered a new instability and extracted its core physical ingredient
- Conducted multifluid dust–gas simulations and revealed implications for planetesimal formation

Tsinghua Univ., Beijing, China

2022–2023

Parametric Inference with Disk Image Prediction Models

ADVISOR: DR. GASPARD DUCHÊNE

- Developed a full physical parameter estimation pipeline for circumstellar disk images
- Designed stochastic processes to quantify correlated image fitting residuals
- Evaluated model uncertainties by high-dimensional posterior calculations with MCMC

Univ. of California, Berkeley, CA

2022–2023

Observational Diagnostics of Outbursting Young Circumstellar Disks (*HL+22*)

ADVISOR: PROF. GREGORY J. HERCZEG

- Built multi-component emission models of viscously heated disks
- Mapped the physical parameter space to disk photometric and spectral features
- Proposed color- and spectral-line-based criteria to search for outbursting disk candidates

Peking Univ., Beijing, China

2021–2022

Publications

- Liu, H.**, Jiang, Y.-F., Quataert, E., Greene, J.E., Ma, Y., Lin, X. 2026, arXiv:2603.02317: *Synthetic Spectral Library of Optically Thick Atmospheres for Little Red Dots*
- Liu, H.**, Jiang, Y.-F., Quataert, E., Greene, J.E., Ma, Y. 2025, ApJ, 994, 113: *The Balmer Break and Optical Continuum of Little Red Dots from Super-Eddington Accretion*
- Liu, H.**, Inayoshi, K., 2025, PRD, 111, 3012: *Gravitational wave forecasts constrained by JWST AGN observations for early massive black hole mergers*
- Liu, H.**, Bai, X.-N. 2023, MNRAS, 526, 80: *The dusty Rossby wave instability (DRWI): linear analysis and simulations of turbulent dust-trapping rings in protoplanetary discs*
- Liu, H.**, Herczeg, G.J., Johnstone, D., et al. 2022, ApJ, 936, 152: *Diagnosing FU Ori-like Sources: The Parameter Space of Viscously Heated Disks in the Optical and Near-infrared*
- Lin, X., Fan, X., Cai, Z., ..., **Liu, H.**, et al. 2026, arXiv:2605.21574: *(LRDs)²: The Low-Redshift Little Red Dots Survey. II. DESI DR1 Sample*
- Chen, Y.-X., **Liu, H.**, Li, R., et al. 2026, arXiv:2602.06954: *Spectral Appearance of Self-gravitating Disks Powered by Stellar Objects: Universal Effective Temperature in the Optical Continuum and Application to Little Red Dots*
- Wang, B., Leja, J., Labbe, I., Greene, J.E., **Liu, H.**, et al. 2026, arXiv:2602.06024: *Water absorption confirms cool atmospheres in two little red dots*
- Hviding, R.E., de Graaff, A., **Liu, H.**, et al. 2026, ApJ, 1000, L18: *The X-Ray Dot: Exotic Dust or a Late-stage Little Red Dot?*
- Lin, X., Fan, X., Cai, Z., Bian, F., **Liu, H.**, et al. 2026, ApJ, 997, 364: *The Discovery of Little Red Dots in the Local Universe: Signatures of Cool Gas Envelopes*
- Greene, J.E., Setton, D.J., Furtak, L.J., ..., **Liu, H.**, et al. 2026, ApJ, 996, 129: *What You See Is What You Get: Empirically Measured Bolometric Luminosities of Little Red Dots*
- Contreras Peña, C., Lee, J.-E., Lee, H.-G., Herczeg, G., Johnstone, D., **Liu, H.**, et al. 2025, ApJ, 987, 23: *“Oh FUors, Where Art Thou?”: A Search for Long-lasting Young Stellar Object Outbursts Hiding in Infrared Surveys*
- Ashraf, M., Jose, J., Lee, H.-G., ..., **Liu, H.**, et al. 2024, MNRAS, 527, 11651: *An Outburst and FU Ori-type Disc of a Former Low Luminosity Protostar*
- Contreras Peña, C., Herczeg, G.J., Ashraf, M., ..., **Liu, H.**, et al. 2023, MNRAS, 521, 5669: *Photometric and spectroscopic monitoring of YSOs in nearby star-forming regions - I. Eruptive YSOs*

Honors & Scholarships

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| 2024 | Peking University Outstanding Graduate , Peking University | |
| 2023 | Outstanding Young Scholar (for research excellence) , Yuanpei College, Peking University | |
| | Yang Jinfang Scholarship for International Exchange , Peking University | 20,000 CNY |
| 2022 | Academic Innovation Award , Peking University | |
| | Merit Student (top 5% in Peking University) , Peking University | |
| | Shu Qi Scholarship for Astronomy & Physics , Peking University | 4,000 CNY |
| 2021 | Pacemaker to Merit Student (top 1% in Peking University) , Peking University | |
| | National Scholarship (for academic excellence) , Ministry of Education of China | 8,000 CNY |

Service, Activities & Development

Teaching assistant: *AST 206 Black Holes* at Princeton University, 2026 Spring

Peer review: *The Astrophysical Journal Letters*; *Astronomy & Astrophysics*

PI for approved JWST program: *A radiative transfer survey for interpreting Little Red Dot spectra* (Cycle 5, AR-10935)